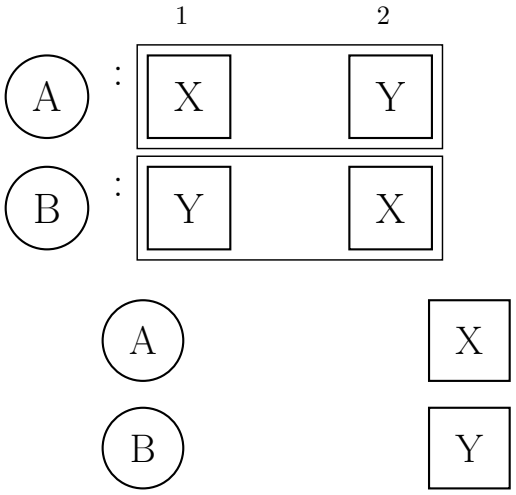
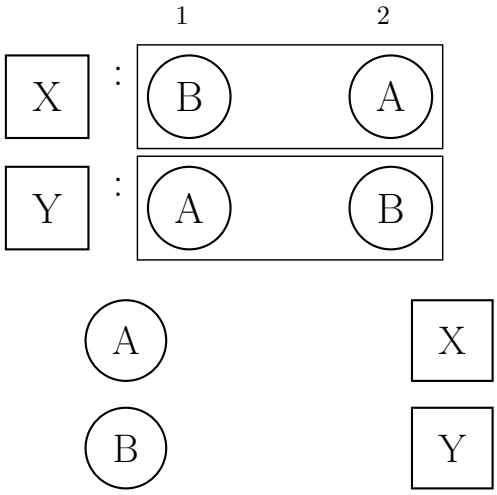
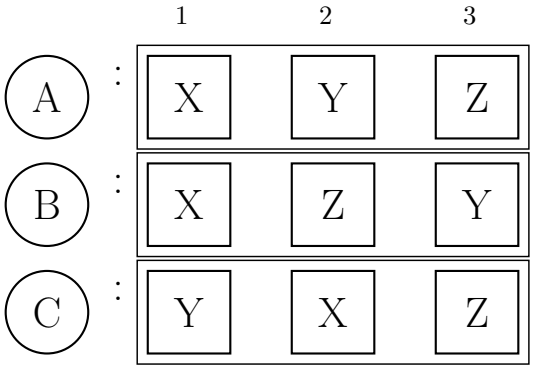
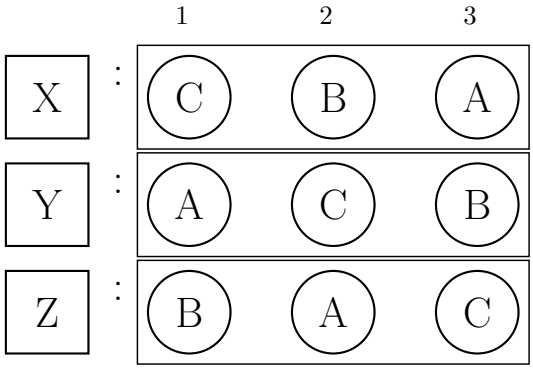
Pair each circle with one square, using every letter once. Each list runs from first choice to last, with no ties. A pairing is stable if no two letters in different pairs both prefer one another to their current partners.

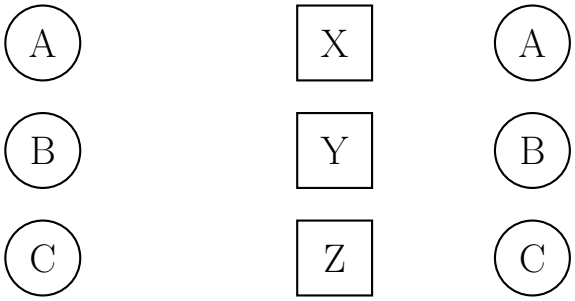
Problem 1: Find all stable pairings for each set of lists.

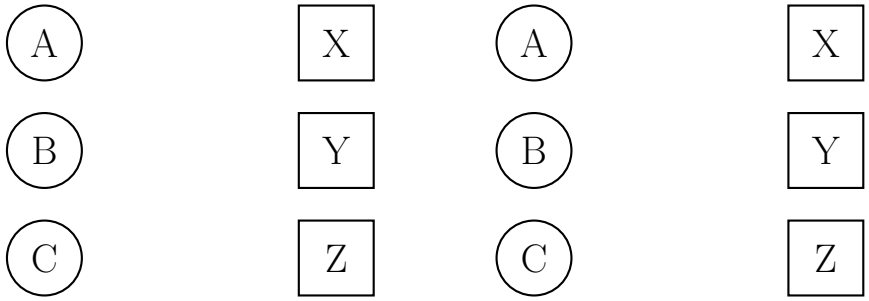








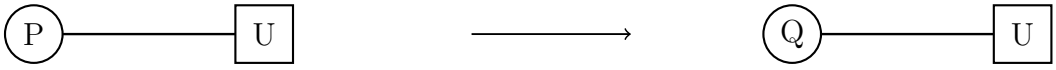




Problem 2: Use these rules with A, B, C, D asking. Show or record the requests and final pairs. Run them again with W, X, Y, Z asking. Does changing the asking side change the final pairing?

Begin with everyone free. A free asker asks its highest choice that it has not asked before. A free receiver holds its asker. A receiver already holding someone keeps whichever of the two it prefers and releases the other. Holds are temporary. Stop when every asker is held.

One step with other letters: U ranks Q ahead of P. Free Q asks U.



Before: P held; Q free. After: Q held; P released.

	1	2	3	4
A	X	W	Z	Y
B	X	Y	W	Z
C	X	Y	Z	W
D	X	Y	W	Z

A, B, C, D asking

	1	2	3	4
W	B	C	D	A
X	D	B	A	C
Y	B	C	A	D
Z	B	D	A	C

W, X, Y, Z asking

